



REVIEW OF RANGE EXTENSIONS OF TROPICAL BRACHYURAN CRABS INTO TEMPERATE WATERS OF SOUTHERN AFRICA

BY

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ABSTRACT

We compiled known occurrence records of seven species of crabs identified in the literature and one documented in this report as having undergone range extensions in southern Africa. Of these eight species, six (*Austruca occidentalis*, *Neosarmatium africanum*, *Ocypode ceratophthalmus*, *Portunus segnis*, *Tubuca urvillei*, and *Varuna litterata*) have extended their ranges either across a major biogeographic boundary that separates the subtropical and warm temperate ecoregions of this coast, or into estuaries that historically lack mangrove forests. For the seventh and eighth species, *Charybdis smithii* and *Scylla serrata*, we were unable to find any distributional data that supported poleward range extension. In addition, we contribute the first record of the blue swimming crab, *P. segnis*, from the temperate south coast of South Africa. This early detection is likely to be part of an ongoing regional trend of tropical-adapted brachyuran crab species extending their ranges into temperate ecoregions.

Key words. — Brachyura, Decapoda, Ocypodidae, Portunidae, range shifts, Sesarmidae, South Africa, Varunidae

RÉSUMÉ

Nous avons compilé les données d'occurrence connues de sept espèces de crabes identifiées dans la littérature et une documentée dans ce rapport comme ayant développé des extensions de répartition en Afrique du Sud. Sur ces huit espèces, six (*Austruca occidentalis*, *Neosarmatium africanum*, *Ocypode ceratophthalmus*, *Portunus segnis*, *Tubuca urvillei*, et *Varuna litterata*) ont étendu leur répartition soit à travers une limite biogéographique majeure qui sépare les écorégions subtropicale et tempérée chaude de cette côte, soit dans les estuaires qui historiquement n'ont pas de forêt de mangrove. Pour les septième et huitième espèces, *Charybdis smithii* et *Scylla serrata*, nous n'avons pas pu trouver de données de distribution qui supportent une extension vers le pôle. De plus, nous apportons le premier signalement du crabe bleu *P. segnis*, de la côte tempérée sud de l'Afrique du Sud. Cette première détection s'inscrit probablement dans une tendance régionale en cours des

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espèces de brachyures adaptées aux tropiques qui étendent leur aire de répartition vers des écorégions tempérées.

Mots clés. — Brachyura, Decapoda, Ocypodidae, Portunidae, changements de répartition, Sesamidae, Afrique du Sud, Varunidae

INTRODUCTION

Four extant marine and estuarine non-native brachyuran crab species (Decapoda, Brachyura) have been documented in South Africa: *Carcinus maenas* (Linnaeus, 1758) in 1983 (Le Roux et al., 1990; Griffiths et al., 2009), *Homalaspis plana* (H. Milne Edwards, 1834) in 2017 (Peters & Robinson, 2018), *Pinnixa occidentalis* Rathbun, 1894 in 2004 (Clark & Griffiths, 2012; Robinson et al., 2020), and *Xantho hydrophilus* (Herbst, 1790) in 2008 (originally misidentified as *Xantho incisus* H. Milne Edwards, 1834; Haupt et al., 2010; Mead et al., 2011; Griffiths et al., 2018). First detected in 1981, one other non-native brachyuran, *Carcinus aestuarii* Nardo, 1847, appears to have gone extinct in South Africa (Geller et al., 1997; Robinson et al., 2009). More recently, Swart & Robinson (2019) have identified at least 28 non-native predatory crab species with potential pathways to South Africa. Among the non-native crabs (i.e., species alien to South Africa) that these authors screened, 27 species have the potential to survive under present-day conditions in the cool temperate waters of the Benguela ecoregion, 27 in the warm temperate waters of the Agulhas ecoregion, 27 in the subtropical waters of the Natal ecoregion, and 26 in the subtropical waters of the Delagoa ecoregion (Swart & Robinson, 2019; fig. 1).

Like species introduced from outside a biogeographic or geopolitical region, extralimital species (i.e., species outside their natural range within a geographic region) can represent a threat to ecological communities, but are often not listed as alien species (Spear & Chown, 2008; Measey et al., 2020). For countries with long and biogeographically varied coastlines, this can be especially concerning when range expansions occur between large biogeographic regions (Emanuel et al., 1992; Spalding et al., 2007) where the extralimital species can affect communities that have evolved independently. Four subtropical and tropical species of marine and estuarine swimming crabs native to South Africa have been introduced elsewhere in the world. These are *Charybdis hellerii* (A. Milne-Edwards, 1867) in the Mediterranean Sea and western Atlantic Ocean (Galil & Zenetos, 2002; Ferry et al., 2017), *Portunus sanguinolentus* (Herbst, 1783) in the Sea of Japan (Kepel & Tsareva, 2005; Spiridonov & Zalota, 2017), *Portunus segnis* (Forskål, 1775) in southern Europe (Lai et al., 2010) and northern Africa (Rabaoui et al., 2015), and *Scylla serrata* (Forskål, 1775) in Hawaii and the western Atlantic Ocean (Eldredge

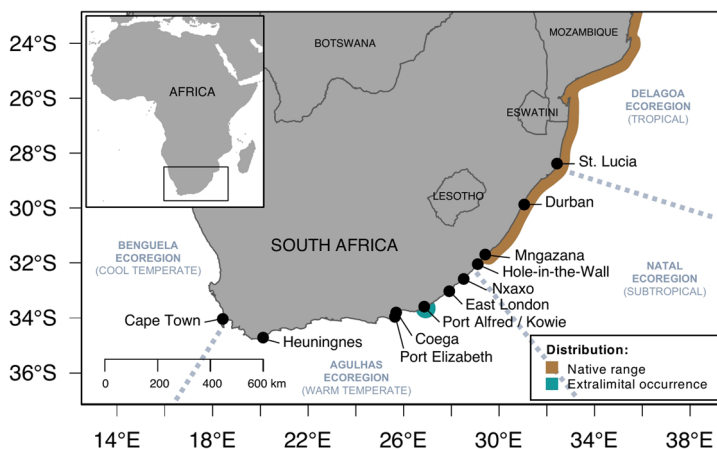


Fig. 1. Native range of *Portunus segnis* (Forskål, 1775) in tropical and subtropical waters of southern Africa, its extralimital occurrence in warm temperate waters, and the four coastal ecoregions of South Africa (ecoregions: Benguela, Agulhas, Natal, and Delagoa) as used in the analysis of Swart & Robinson (2019).

& Smith, 2001; Tavares & Mendonça, 2011; Lemaitre et al., 2013; Metri et al., 2020). In South Africa, the historical distributions of these four predatory crab species are confined to the east and southeast coasts (Day, 1967; Branch et al., 2010). As extralimital species, they have the potential to breach biogeographic boundaries and spread westwards along the coast of southern Africa. For instance, elsewhere in the world, crab species have been shown to expand their range by 100s of km in response to environmental change, such as marine heatwaves (Lonhart et al., 2019).

In this paper, we evaluated the native distribution and purported range extensions leading to extralimital occurrences of tropical brachyuran crabs in southern Africa. These putative range-extending species, comprised *Austruca occidentalis* (Naderloo, Schubart & H.-T. Shih, 2016), *Charybdis smithii* MacLeay, 1838, *Neosarmatium africanum* Ragionieri, Fratini & Schubart, 2012, *Ocypode ceratophthalmus* (Pallas, 1772), *S. serrata*, *Tubuca urvillei* (H. Milne Edwards, 1852), and *Varuna litterata* (Fabricius, 1798), all of which were believed to be spreading from subtropical and tropical waters (east coast) to temperate waters (west and/or south coasts) of South Africa (Branch, 1984; Whitfield et al., 2016; Kajee et al., 2018; Peer et al., 2018a, b; Adams et al., 2019). In addition, we document the first extralimital record of the tropical blue swimming crab, *P. segnis*, in the warm temperate Agulhas ecoregion of the south coast of South Africa (i.e., outside of the species' known subtropical/tropical range on the east coast) and discuss its superficial resemblance to another blue swimming crab, *Callinectes sapidus* Rathbun, 1896, which, unlike *P. segnis*, is a species on the South African watch list of alien crabs (Swart & Robinson, 2019).

MATERIAL AND METHODS

Review of range-extending crabs

A comprehensive literature search was conducted for occurrence records of *Austruca occidentalis*, *Charybdis smithii*, *Neosarmatium africanum*, *Ocypode ceratophthalmus*, *Portunus segnis*, *Scylla serrata*, *Tubuca urvillei*, and *Varuna litterata* (and their synonyms as used in the region) in southern Africa (table I). We used records found in the published literature (including technical reports and student theses) and the Ocean Biodiversity Information System (OBIS; <https://obis.org/>). For each of the eight target species, we assessed the distribution of presence and non-presence records to categorize records as within the species' native range (i.e., intralimital species) or outside of its native range (i.e., extralimital species). As most of the target species are mangrove-associated, we used two approaches to access their distribution. First, we used the historical geographic limits of mangroves at the Nxaxo Estuary (Macnae 1963; Peer et al., 2018a; fig. 1) as proxy for the habitat characterizing tropical and subtropical estuaries. Detection of a species west of this estuary was regarded as an extralimital record. Second, non-presence records from baseline surveys of estuaries that were not associated with mangrove forests were further used to support the determination of extralimital records (Day et al., 1951; Scott et al., 1951; Millard & Scott, 1954; Brown & Day, 1959; Branch & Day, 1984; Hodgson, 1987). Finally, we consulted the World Register of Introduced Marine Species (WRiMS, 2021) to determine the status of each target species as an introduced species at a global level.

Portunus segnis (Forskål, 1775)

Material examined. — A single, young, male specimen of *Portunus segnis* (fig. 2) was caught by a local recreational fisher in the morning of 28 January 2020 with a fishing net after it was spotted swimming in the surface waters of the Kowie River (about 4 km upstream from the river mouth), Port Alfred, South Africa (33°35'18.9"S 26°52'07.0"E; fig. 1).

Descriptive notes.— The carapace was wider (132 mm, including lateral spines) than long. The specimen exhibited nine blunt teeth (including a strong lateral spine) on the arched anterolateral margin. The number and morphology of frontal teeth could not be resolved from images. The dorsal surface of the carapace was olive green in colour with fine granules and without any spot patterns. Mottled patterns (lighter in colouration) were present on the chelipeds. A mesiodistal spine was present on the carpus of the cheliped, which is a morphological feature that distinguishes the genus *Portunus* from the genus *Callinectes* (Tavares, 2002). A sharp spine on the inner angle of the carpus of the cheliped and three spines on the anterior margin of the merus of the cheliped were present. The fifth legs were

TABLE I

Review of eight putative range-extending brachyuran crab species that are native to southern Africa: *Austruca occidentalis* (Naderloo, Schubart & H.-T. Shih, 2016), *Charybdis smithii* MacLeay, 1838, *Neosarmatium africanum* Ragionieri, Fratini & Schubart, 2012, *Ocypode ceratophthalmus* (Pallas, 1772), *Portunus segnis* (Forskål, 1775), *Scylla serrata* (Forskål, 1775), *Tubuca urvillei* (H. Milne Edwards, 1852), and *Varuna litterata* (Fabricius, 1798)

Species (synonyms used in southern Africa)	Native range in southern Africa (sources of occurrence records)	Extralimital occurrence and distance from native range	Status outside of southern Africa
<i>Austruca occidentalis</i> (<i>Austruca annulipes</i> , <i>Gelasimus lacteus</i> , <i>Uca annulipes</i> , and <i>Uca lactea</i> var. <i>annulipes</i> , <i>Uca lactea</i> ; also see Naderloo et al., 2016)	Warm temperate to tropical: south and east coasts, southern limit at Nxaxo (Krauss, 1843; Stebbing, 1910; Boyce, 1924; Barnard, 1950; Day et al., 1954; Day & Morgans, 1956; Broekhuysen & Taylor, 1959; Macnae, 1962, 1963; Crosnier, 1965; Day, 1969, 1974, 1981; Kensley, 1981; Emmerson, 1994; de Boer & Longamane, 1996; de Boer & Prins, 2002; Litulo, 2005a; Vorsatz, 2009; Silva et al., 2010; Peer et al., 2014, 2018a, b; Emmerson, 2016; OBIS, 2021a)	Warm temperate: Nahoon, 80 km (Peer et al., 2018a); Kariëga, 235 km (Hodgson, 1987; Peer et al., 2018a); Knysna, 695 km (Peer et al., 2015, 2018a)	Not known to be an introduced species (WRiMS, 2021)
<i>Charybdis smithii</i> (<i>Gonioneptunus smithii</i> , <i>Goniosoma truncatum</i> ; also see Balasubramanian, 1993)	Cool temperate to tropical: west, south, and east coasts, westernmost limit at Blouberg (MacLeay, 1838; Milne-Edwards, 1861; Krauss, 1843; Stebbing, 1910; Barnard, 1950; Kensley, 1977, 1981; Branch 1984; Balasubramanian, 1993; van Couwelaar et al., 1997; Emmerson, 2016; OBIS, 2021b; G. M. Branch, pers. comm.)	No extralimital records because observations of strandings on beaches were assumed to represent catastrophic events (e.g., Branch, 1984; Romanov et al., 2015)	Not known to be an introduced species (WRiMS, 2021)

TABLE I
(Continued)

Species (synonyms used in southern Africa)	Native range in southern Africa (sources of occurrence records)	Extralimital occurrence and distance from native range	Status outside of southern Africa
<i>Neosarmatium africanum</i> (<i>Neosarmatium meinerti</i> , <i>Sesarma meinerti</i> , <i>Sesarma tetragona</i> , <i>Sesarma tetragonum</i> ; also see Ragionieri et al., 2012)	Warm temperate to tropical: south and east coasts, southern limit at Nxaxo (Krauss, 1843; Stebbing, 1910; Barnard, 1950, 1955; Day et al., 1954; Day & Morgans, 1956; Broekhuysen & Taylor, 1959; Macnae, 1962, 1963; Macnae & Kalk, 1962a, b; Crosnier, 1965; Day, 1969, 1974, 1981; Kensley, 1981; Millard & Harrison, 1984; Steinke et al., 1993; Wiseman et al., 1993; Emmerson, 1994, 2016; Vorsatz, 2009; Peer et al., 2014, 2018a; Fusi et al., 2017; OBIS, 2021c)	Warm temperate: Nahoon, 80 km (Peer et al., 2018a); Tyolomnqa, 130 km (Peer et al., 2018a); Kariega, 235 km (Hodgson, 1987)	Not known to be an introduced species (WRiMS, 2021)
<i>Ocypode ceratophthalminus</i> (as <i>Ocypode ?ryderi</i> in Branch, 1984; sensu Kajee et al., 2018)	Subtropical and tropical: east coast, southern limit at Mngazana (Krauss, 1843; Stebbing, 1910; Barnard, 1950; Day et al., 1954; Day & Morgans, 1956; Broekhuysen & Taylor, 1959; Macnae & Kalk, 1962b; Macnae, 1963; Crosnier, 1965; Day, 1969, 1974; Kensley, 1970, 1981; McLachlan, 1980; Millard & Harrison, 1984; Clarke, 1989; Jackson et al., 1991; de Boer & Longamane, 1996; de Boer & Prins, 2002; Vorsatz, 2009; Peer et al., 2014; Emmerson, 2016; OBIS, 2021d)	Warm temperate: Port Elizabeth, 500 km (Barnard, 1950); Mossel Bay, 960 km (Barnard, 1950; Kensley, 1981; Jackson et al., 1991; OBIS, 2021c); Palmiet, 1400 km (Branch, 1984; Branch & Day, 1984)	Not known to be an introduced species (WRiMS, 2021)

TABLE I
(Continued)

Species (synonyms used in southern Africa)	Native range in southern Africa (sources of occurrence records)	Extralimital occurrence and distance from native range	Status outside of southern Africa
<i>Portunus segnis</i> (<i>Lupa pelagica</i> , <i>Lupa pubescens</i> , <i>Neptunus pelagicus</i> , <i>Portunus pelagica</i> , <i>Portunus pelagicus</i> ; also see Lai et al., 2010)	Subtropical to tropical: east coast, southern limit at Mgazana (Krauss, 1843; Stebbing, 1921; Barnard, 1950; Day et al., 1954; Day & Morgans, 1956; Crosnier, 1962; Macnae & Kalk, 1962b; Day, 1969, 1974; Branch & Grindley, 1979; Kensley, 1981; de Boer & Longamane, 1996; de Boer & Prins, 2002; Peer et al., 2014; Emmerson, 2016; OBIS, 2021e)	Warm temperate: Port Alfred/ Kowie, 370 km (this report)	Introduced to the Mediterranean Sea (Fox, 1924; Barnard, 1950; Crocetta, 2006; Yokes et al., 2007; Lai et al., 2010; Rabaoui et al., 2015; Emmerson, 2016)
<i>Scylla serrata</i> (<i>Achelous crassimanus</i>)	Warm temperate to tropical: south and east coasts, southern limit at Heuningnes (Krauss, 1843; Stebbing, 1910; Barnard, 1950; Day et al., 1951, 1954; Day & Morgans, 1956; Broekhuysen & Taylor, 1959; Macnae, 1957, 1963; Crosnier, 1962; Macnae & Kalk, 1962a; Day, 1969, 1974, 1981; Hill, 1975, 1979; Heinecken, 1981; Heinecken & Grindley, 1982; Kensley, 1981; Bickerton, 1984; Duvenage & Morant, 1984; Millard & Harrison, 1984; Baird et al., 1986; Hodgson, 1987; Burns et al., 1988; Forbes & Hay, 1988; Clarke & Brownlie, 1990; Robertson & Piper, 1991; Wiseman et al., 1993; de Boer & Longamane, 1996; Robertson, 1996; Gopurenko et al., 1999; Cyrus, 2001; de Boer & Prins, 2002; Vorsatz, 2009; Emmerson, 2016; Peer et al., 2014, 2018a; Rumisha et al., 2017; OBIS, 2021f)	No extralimital records because baseline surveys suggest continuous distribution on the south coast	Introduced to the Caribbean Sea (Lemaître et al., 2013), western Atlantic (Tavares & Mendonça, 2011; Metri et al., 2020), the Malaysian waters (cryptogenic species; Fazhan et al., 2017), and the Hawaiian waters (Edmondson, 1954; Emmerson, 2016; Hurley et al., 2020)

TABLE I
(Continued)

Species (synonyms used in southern Africa)	Native range in southern Africa (sources of occurrence records)	Extralimital occurrence and distance from native range	Status outside of southern Africa
<i>Tubuca urvillei</i> (<i>Gelasimus arcuatus</i> , <i>Uca arcuata</i> , <i>Uca urvillei</i>)	Warm temperate to tropical: south and east coasts, southern limit at Nxaxo (Krauss, 1843; Stebbing, 1905, 1910; Barnard, 1950; Day et al., 1954; Day & Morgans, 1956; Macnae, 1962, 1963; Macnae & Kalk, 1962a; Crosnier, 1965; Day, 1969, 1974, 1981; Kensley, 1981; Emmerson, 1994; Litulo, 2005b; Vorsatz, 2009; Peer et al., 2014, 2018a, b; Emmerson, 2016; OBIS, 2021g)	Warm temperate: Nahoon, 80 km (Peer et al., 2018a); Kariega, 235 km (Hodgson, 1987; Peer et al., 2015, 2018a)	Not known to be an introduced species (WRiMS, 2021)
<i>Varuna litterata</i> (<i>Varuna tomentosa</i> ; also see Ng et al., 2008)	Subtropical to tropical: east coast, southern limit at Hole-in-the-Wall (Stebbing, 1905, 1910; Barnard, 1950, 1955; Day et al., 1954; Day & Morgans, 1956; Crosnier, 1965; Day, 1974, 1981; Bruton & Appleton, 1975; Morant & Grindley, 1982; Blaber et al., 1984; Millard & Harrison, 1984; Connell & Robertson, 1986; Cyrus, 2001; Vorsatz, 2009; Peer et al., 2014, 2018a; Emmerson, 2016; OBIS, 2021h; G. M. Branch, pers. comm.)	Warm temperate: Swartkops, 440 km (Macnae, 1957; N. Peer, pers. comm.); Breë, 1060 km (Kensley, 1981); Palmiet, 1340 km (Kajee et al., 2018); Muizenberg/Sandvlei, 1450 km (Morant & Grindley, 1982; OBIS, 2021h)	Not known to be an introduced species (WRiMS, 2021), but poleward range extension in Australia was documented (Mos et al., 2017)

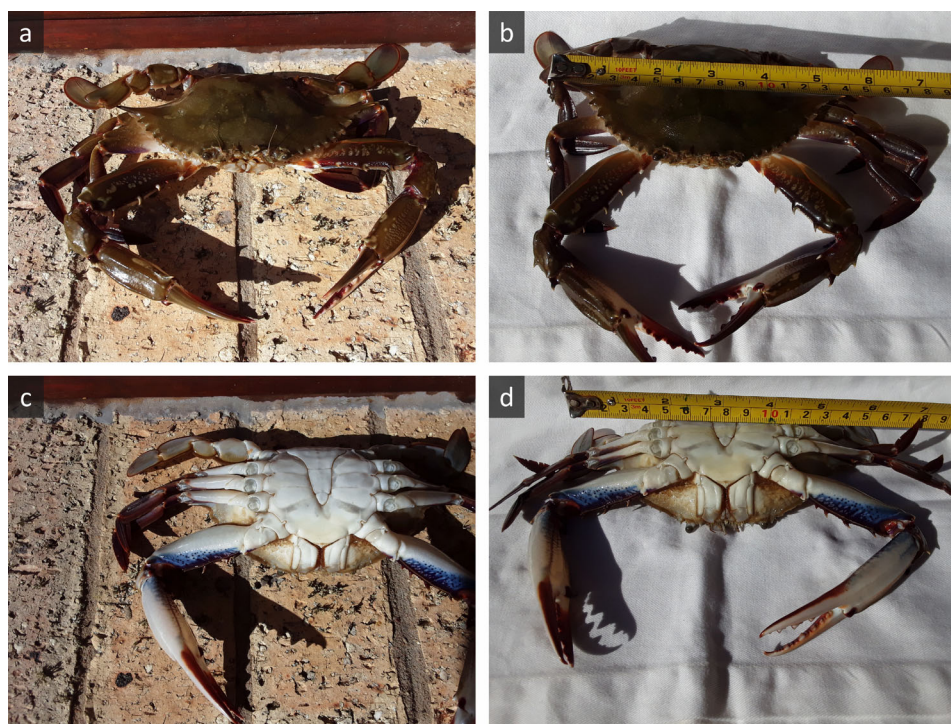


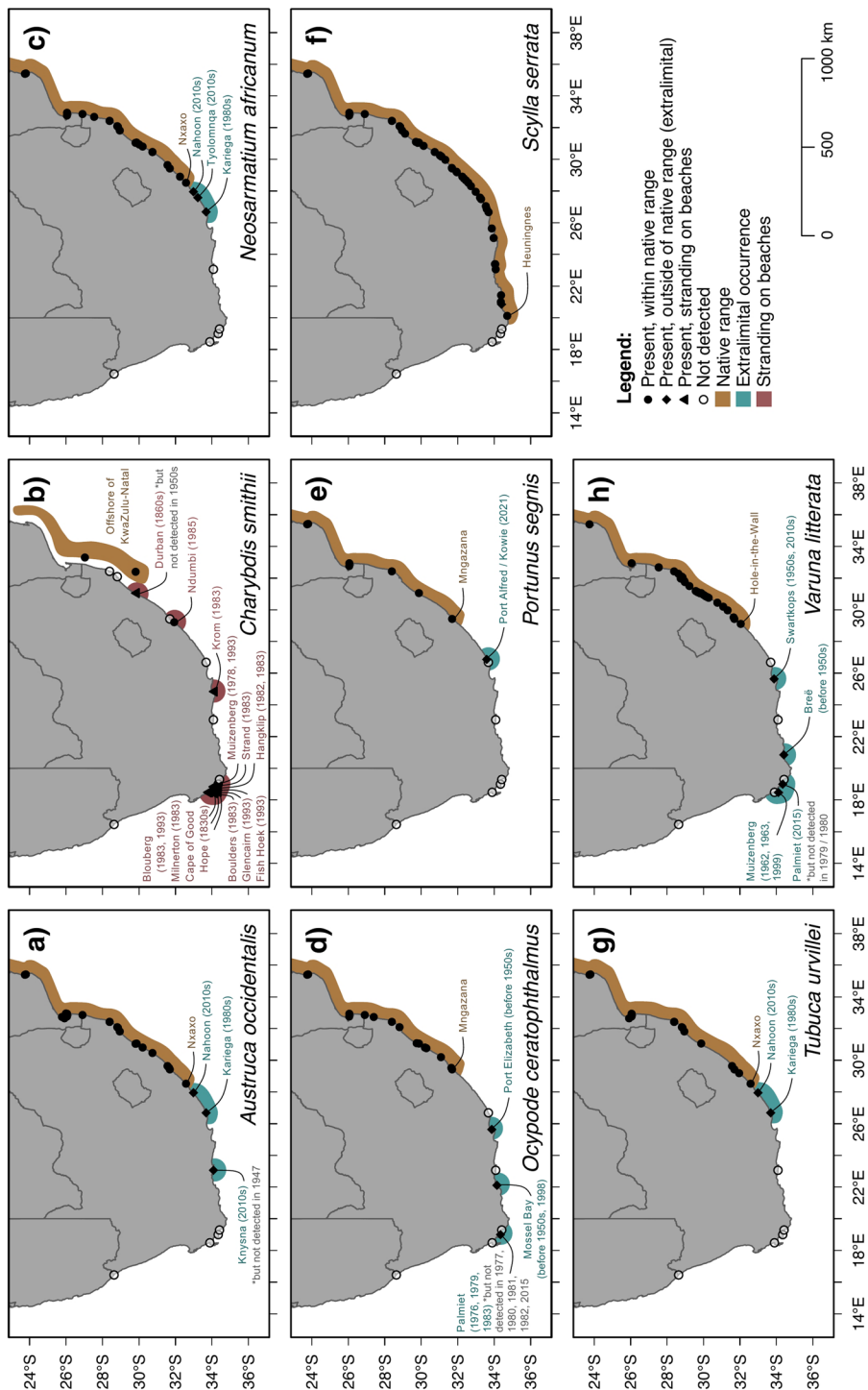
Fig. 2. A single, young, male *Portunus segnis* (Forskål, 1775) found in the Kowie River, Eastern Cape, South Africa: a–b, oblique dorsal view; c–d, oblique ventral view. Photos were taken ex situ by a recreational fisher and used with permission. Scale provided in panels (b) and (d). [Photos by R. De Cauwer.].

flattened. The joints of the legs exhibited tinges of red. The abdomen was triangle-shaped and reached the thoracic sternite 4 (Crocetta, 2006; Lai et al., 2010). The third, fourth, and fifth somites of the abdomen were fused (Crosnier, 1962). The sixth somite of the abdomen was wider than the telson and widened distally. The inner surface of the manus of the chelae was blue. The underside was white in colour.

RESULTS AND DISCUSSION

Review of range-extending crabs

In southern Africa, prior to the discovery of the extralimital occurrence of *Portunus segnis* (this report), seven other tropical brachyuran crab species were believed to have exhibited range extensions from the east coast (subtropical and tropical waters) to the south coast (warm temperate waters; Branch, 1984; Whitfield et al., 2016; Kajee et al., 2018; Peer et al., 2018a, b; Adams et al., 2019).



Our review of historical presence records of these crab species throughout southern Africa (geographic coverage: east and south coasts of South Africa and southern Mozambique) and their non-presence records (west and south coasts of South Africa) supported range extensions for *Austruca occidentalis*, *Neosarmatium africanum*, *Ocypode ceratophthalmus*, *P. segnis*, *Tubuca urvillei*, and *Varuna litterata*, but not for *Charybdis smithii* or *Scylla serrata*.

The southernmost limit of the native range of *O. ceratophthalmus*, *P. segnis* and *V. litterata* extends to the Transkei region (either Hole-in-the-Wall or the Mngazana Estuary), which is situated well within the subtropical waters of the Natal ecoregion (table I; figs. 1 and 3). By contrast, the southern limits of the native ranges of *A. occidentalis*, *N. africanum*, and *T. urvillei* extend to the Nxaxo Estuary (i.e., the historical southern limit of mangrove forests in southern Africa; Macnae 1963; Peer et al., 2018a), which is located in the warm temperate waters of the Agulhas ecoregion, but close (about 60 km) to the transition zone between the Agulhas and Natal ecoregions (table I; figs. 1 and 3). The range-edge of *S. serrata* is at the Heuningnes River Estuary (Day, 1981; Bickerton, 1984; Robertson, 1996), which is further west (about 1120 km) from this biogeographic boundary (table I; figs. 1 and 3).

A baseline survey of the Kariëga Estuary in the 1980s and more recent work detected three extralimital crab species — *A. occidentalis* (as *Uca annulipes* (H. Milne Edwards, 1837)), *N. africanum* (as *Sesarma meinerti* De Man, 1887), and *Tubuca urvillei* (as *Uca urvillei* (H. Milne Edwards, 1852)) — that occurred beyond the historical geographic limits of mangrove ecosystems in southern Africa, i.e., south of the Nxaxo Estuary (Hodgson, 1987; Peer et al., 2015, 2018a; table I; figs. 1 and 3). Notably, one of the more recent extralimital observations of *A. occidentalis* from Knysna Lagoon in the 2010s (Peer et al., 2015, 2018a) came from a locality where the species was not detected when surveyed in 1947 (Day et al., 1951).

Baseline surveys done in estuaries on the south and west coasts of South Africa (Day et al., 1951; Scott et al., 1951; Millard & Scott, 1954; Brown & Day,

Fig. 3. Native and extralimital distributions of eight putative range-extending brachyuran crab species in southern Africa [one species documented in this report and seven other species identified by Branch (1984), Whitfield et al. (2016), and Kajee et al. (2019)]: a, *Austruca occidentalis* (Naderloo, Schubart & H.-T. Shih, 2016); b, *Charybdis smithii* MacLeay, 1838; c, *Neosarmatium africanum* Ragionieri, Fratini & Schubart, 2012; d, *Ocypode ceratophthalmus* (Pallas, 1772); e, *Portunus segnis* (Forskål, 1775); f, *Scylla serrata* (Forskål, 1775); g, *Tubuca urvillei* (H. Milne Edwards, 1852); and h, *Varuna litterata* (Fabricius, 1798). Sources of presence records are provided in table I and non-presence records (i.e., species not detected) were based on surveys of estuaries done by Day et al. (1951, 1954), Scott et al. (1951), Millard & Scott (1954), Day & Morgans (1956), Brown & Day (1959), Millard & Broekhuysen (1970), Branch & Grindley (1979), Branch & Day (1984), Millard & Harrison (1984), and Hodgson (1987).

1959; Branch & Day, 1984; Hodgson, 1987) indicated that *O. ceratophthalmus*, *P. segnis*, and *V. litterata* were historically absent from these estuarine systems (except for the detection of *O. ceratophthalmus* from the Palmiet Estuary; fig. 3). This suggests that observations of these three species in temperate waters of the Benguela and Agulhas ecoregions can be considered extralimital occurrences. To date, there are: one extralimital record of *P. segnis* (from the Kowie River in 2021; this report), six records of *O. ceratophthalmus* (from Mossel Bay and Port Elizabeth before the 1950s, the Palmiet Estuary in 1976, 1979 and 1983, and Mossel Bay in 1998; Barnard, 1950; Branch, 1984; Branch & Day, 1984; OBIS, 2021d), seven records of *V. litterata* (from the Swartkops Estuary in the 1950s, the Muizenberg/Sandvlei Estuary in 1962, 1963, and 1999, Hole-in-the-Wall in 2017, the Breë River before the 1980s, and the Palmiet Estuary in 2015; Macnae, 1957; Kensley, 1981; Emmerson, 2016; Kajee et al., 2018; OBIS, 2021h; fig. 3). In this review, we assumed that the specimen identified as *Ocypode ?ryderi* Kingsley, 1880 (cf. Branch, 1984) from the Palmiet Estuary was, in fact, *O. ceratophthalmus* (sensu Kajee et al., 2018), because Branch (1984) and Branch & Day (1984) both reported low numbers of *Ocypode* juveniles from this locality in 1979 (one individual per 100 m of shoreline and ‘rare’, respectively). Besides the shared locality and year of observation, only one species in the genus *Ocypode* was reported in each source, suggesting the two sources reported the same species. The discovery of *V. litterata* in the Palmiet Estuary in 2015 was notable because it was not detected in this estuary when it was surveyed in 1979/1980 (Branch & Day, 1984). In addition to the above records, there are two unpublished records of *V. litterata* — an extralimital detection from Swartkops Estuary in the 2010s (N. Peer, pers. comm.) and an intralimital detection of a few adults and a number of megalopa larvae from Hole-in-the-Wall on 25 March 2017 (G. M. Branch, pers. comm.). For both *O. ceratophthalmus* and *V. litterata*, their extralimital records occurred sporadically in warm temperate waters across an extensive distance on the south coast of South Africa, but nonetheless within the limits of the Agulhas ecoregion. These extralimital populations may be highly ephemeral; for instance, *O. ceratophthalmus* appeared to have disappeared from the Palmiet Estuary by 2015 (Kajee et al., 2018).

Charybdis smithii is a pelagic and deep-water benthic tropical species that forms dense swarms nearshore during the larval phases of its life cycle (Kensley, 1977, 1981; Balasubramanian, 1993; van Couwelaar et al., 1997). Despite being a tropical species, it has been found stranded on beaches on the south and west coasts of South Africa (MacLeay, 1838; Branch, 1984; OBIS, 2021b; fig. 3). These strandings in temperate waters appeared to be linked to years with anomalously elevated water temperatures (Branch, 1984). In addition to the stranding of adult *C. smithii*, strandings of megalopa larvae were observed on beaches of both Table

Bay and False Bay around Cape Town in 2018 (G. M. Branch, pers. comm.). As compelling as these coastal occurrences of *C. smithii* are, in this review, we did not consider observations of stranded animals as extralimital records despite being far afield from their native range, because these individuals will ultimately die, as strandings were usually catastrophic events (Branch, 1984; Romanov et al., 2015) and because there is no direct evidence yet of an extralimital population in offshore temperate waters.

No extralimital records of *Scylla serrata* were found in this review because this species has been documented in several baseline faunal surveys done in the warm temperate waters of the south coast of South Africa (Day et al., 1951; Hodgson, 1987) and from other studies done in estuaries outside the geographic limits of mangrove ecosystems (Barnard, 1950; Hill, 1975; fig. 3). The presence of populations across its southern African range, coupled by the lack of non-presence records from baseline surveys of estuaries on the south coast, suggests that its distribution on the south coast is well within its native range (fig. 3). The lack of evidence for a poleward range extension in *S. serrata* may be because this species did not undergo contemporary range shifts in southern Africa or because its spread occurred prior to sufficient baseline information.

Records of extralimital occurrences of *O. ceratophthalmus*, *P. segnis*, and *V. literata* exhibited a large gap of between 370 and 500 km from the edge of their native range, while the gap for *A. occidentalis*, *N. africanum*, and *T. urvillei* was smaller, about 80 km (table I). Although not explicitly stated in the literature as undergoing range extensions in southern Africa, two additional tropical species have been sighted outside their purported native ranges: *Ocypode cursor* (Linnaeus, 1758) from the Orange River Mouth (28°38'04.3"S 16°27'14.9"E) on 14 September 2017 (G. M. Branch, pers. comm.) and *Paraleptuca chlorophthalmus* (H. Milne Edwards, 1837) from the Kariega Estuary (33°40'49.3"S 26°40'55.8"E) in March 2016 (Peer et al., 2018a) and the adjacent Bushman's River Estuary (33°41'12.5"S 26°39'43.6"E) in 2021 (N. Peer, pers. comm.). In the southern African context, range extensions of many of the brachyuran crab species from their native range in subtropical and tropical waters to warm temperate waters have been attributed to climate warming, poleward expansion of mangrove distribution, and/or once-off anomalous warm temperature events (Branch, 1984; Whitfield et al., 2016; Peer et al., 2018a, b; Adam et al., 2019); while, of the crab species examined in this review, only *Portunus segnis* and *S. serrata* are known to have been introduced to other ecoregions of the world (WRiMS, 2021; table I).

Extralimital record of *Portunus segnis*

Portunus segnis was originally described from the Red Sea and is now considered native to the western Indian Ocean, from Pakistan to the east coast of South

Africa (Lai et al., 2010; Emmerson, 2016). By 1898, *P. segnis* had been introduced to the Mediterranean Sea after the opening of the Suez Canal (Fox, 1924; Barnard, 1950; Lai et al., 2010; Emmerson, 2016). At present, the species has been sighted throughout the Mediterranean Sea in the Levantine Sea, Israel (Fox, 1924), the Tyrrhenian Sea, Italy (Crocetta, 2006), the Aegean coast of Turkey (Yokes et al., 2007), and the Gulf of Gabes, Tunisia (Rabaoui et al., 2015). Observations of the species previously reported it as *P. pelagicus* (Linnaeus, 1758) sensu lato until the species complex of four similar species was revised and specimens from its native and introduced range in the western Indian Ocean and the Mediterranean Sea were assigned to *P. segnis* (cf. Lai et al., 2010; Shaiek et al., 2021). The species inhabits marine and estuarine waters over mud or sand and is associated with mangroves in its native range (Carpenter et al., 1997; Safaie et al., 2013; Emmerson, 2016). Where it is alien in the Mediterranean Sea, however, *P. segnis* tends to appear under rocks and in intertidal rock pools associated with muddy or sandy bottoms and is only sometimes sighted in estuaries (Galil et al., 2002; Lai et al., 2010; Shaiek et al., 2021).

Although our specimen superficially resembles the alien *Callinectes sapidus*, we determined its identity to be either *Portunus pelagicus* sensu stricto or *Portunus segnis* based on the presence of the mesiodistal spine on the carpus of the cheliped (absent in *C. sapidus*) and the triangle-shaped male abdomen (T-shaped in *C. sapidus*; fig. 2). Further, we deduced that the specimen was *P. segnis* and not *P. pelagicus* sensu stricto based on the known range delimitation of *P. segnis* and its occurrence as an alien species elsewhere (Lai et al., 2010; Emmerson, 2016; Bagheri et al., 2020). To date, there are no records of *P. pelagicus* sensu stricto having been introduced to areas outside its natural range (Lai et al., 2010). Our determination is, however, based on morphological identification via photographs because there was no physical material to confirm its identity using molecular approaches.

An Italian record initially misidentified *P. segnis* (referred to as *P. pelagicus* sensu lato) as *C. sapidus*, but was later corrected (Crocetta, 2006). More recently, Cerri et al. (2020) have suggested that a record of *C. sapidus* from the Ionian coast of Sicily may again have been a misidentification of *P. segnis*. Such errors highlight the importance of avoiding the misidentification of alien species, including portunid crabs.

In the case of *Portunus* spp., a high degree of variation in colour patterns probably contributes to misidentifications. In particular, colour patterns of *P. segnis* vary with the associated habitat substratum type, presumably as camouflage against visual predators (Giraldes et al., 2016). The red tinged finger tips of the cheliped exhibited by our specimen is typical of females (Lai et al., 2010), which is consistent with the observation that young males can present female features as

a strategy to avoid competition (Giraldes et al., 2016). Indeed, the carapace and cheliped colour patterns of our young male specimen of *P. segnis* closely resembled those of a female *P. pelagicus* sensu stricto specimen from Indonesia photographed by Fujaya et al. (2016).

The *P. pelagicus* species complex consists of four cryptic species, including *P. segnis* (cf. Ng et al., 2008; Lai et al., 2010). Based on the presence of three spines on the anterior margin of the merus of the cheliped and the relatively shorter and broader sixth somite of the male abdomen, we can narrow the candidate species to *P. pelagicus* sensu stricto and *P. segnis* (cf. Lai et al., 2010). These species are genetically closely related, diverging about 1.5 to 2.5 million years ago, which is also reflected in their morphological similarities (Lai et al., 2010).

Based on the known historical range, Lai et al. (2010) considered the distribution of *P. pelagicus* sensu stricto to extend from Kyushu in Japan as far west as the Bay of Bengal, without geographically overlapping with *P. segnis*, which is found on the west of the Indian sub-continent. More recently, Bagheri et al. (2020), however, showed that *P. pelagicus* sensu stricto is present at low abundances as far west as the Persian Gulf and the Oman Sea, where *P. pelagicus* sensu stricto and *P. segnis* co-exist. Within its known range in South Africa, *P. segnis* (variously described as *Lupa pelagica* (Linnaeus, 1758), *Lupa pubescens* Dana, 1852, *Neptunus pelagicus* (Linnaeus, 1758), *Portunus pelagica* (Linnaeus, 1758), and *P. pelagicus* sensu lato) has been reported from Mngazana Estuary (Branch & Grindley, 1979), Durban (Krauss, 1843; Stebbing, 1921; Barnard, 1950; Day & Morgans, 1956; Day, 1969), and St. Lucia (Day et al., 1954; Millard & Broekhuysen, 1970; Peer et al., 2014), and is considered to be confined to the subtropical and tropical waters of the Natal and Delagoa ecoregions (Crosnier, 1962; Kensley, 1981; Lai et al., 2010; Bagheri et al., 2020; figs. 1 and 3).

In South Africa, *P. pelagicus* sensu lato and other crab species have been sold and consumed in Cape Town and Durban and their environs (Cawthorn & Hoffman, 2017). On the South African market, these products were labelled 'blue crab' or 'blue swimming crab' and were sold dead and in frozen form (D. M. Cawthorn, pers. comm.). Of the products tested by Cawthorn & Hoffman (2017), one product was listed as originating from Mozambique and the other as having been processed in South Africa (D. M. Cawthorn, pers. comm.). This accords with the existence of a fishery for *P. segnis* (reported as *P. pelagicus* sensu lato) in Mozambique (de Boer & Longamane, 1996; de Boer & Prins, 2002; Machava et al., 2014). In South Africa, crab species may be used as bait by recreational fishers, but crabs used as bait tend to be locally-found, small-sized species, which does not match the profile of *P. segnis*. Anecdotally, the recreational fisher who captured the *P. segnis* specimen described here has subsequently only caught the native mud crab, *Scylla serrata* (Forskål, 1775), which is common in the Kowie River.

The dispersal ability of *P. segnis* larvae is poorly understood, with most studies done on *P. pelagicus* sensu lato and outside the distributional range of *P. segnis* (e.g., Bryars & Havenhand, 2004). Significant genetic structure among some of the surveyed populations of *P. segnis* in Iran suggests that either larval dispersal is relatively limited at the 100s km scale, or that physical barriers (e.g., hydrodynamics) mediate horizontal larval transport (Bagheri et al., 2020). Larval periods of 26 to 45 days of the closely related species, *P. pelagicus* sensu lato, in temperate waters of Australia (Bryars & Havenhand, 2006) probably correspond to only 10s km in realized dispersal distances (i.e., range: 33 to 160 km) based on data for other brachyuran species with similar larval periods (Shanks et al., 2003). As an introduced species in the Mediterranean Sea, the main vectors associated with the spread of *P. segnis* are unknown but are suspected to involve shipping activities (Manfrin et al., 2018). In particular, decapod larvae are presumed to survive well in ballast water (Ambrogi, 2000).

The circumstances surrounding the appearance of *P. segnis* into the Kowie River are unknown. Given, however, that the Kowie River mouth (Port Alfred) is about 370 km along-shore outside its native range (fig. 1), we suspect that the species was unintentionally introduced via a pathway associated with shipping activities (e.g., offshore ballast water discharge or at a nearby port). Despite its name, Port Alfred is not a functional port, though there is a recreational marina near the river mouth. It is situated between three busy ports: a river port in East London to the east and two coastal ports at Coega and Port Elizabeth to the west (fig. 1). East London is about 120 km (by sea) to the east of Port Alfred, while Coega and Port Elizabeth are about 120 and 125 km, respectively, to the west. The possibility that the species was intentionally released is unlikely, given that crab products are typically sold frozen at markets and other native crab species are more likely to be used as bait by local recreational fishers. Of course, it is plausible that transient recreational fishers, visiting during the holiday season, may have taken *P. segnis* out of its native range and to the Kowie River. Transient and resident (and, occasionally, newly launched) yachts and small crafts are typically observed in Port Alfred throughout the year. During a roughly monthly census between August 2020 and March 2021, we counted 96 (10 August), 87 (10 September), 91 (12 October), 94 (21 October), 100 (12 November), 135 (17 December), 130 (11 January), 112 (8 February), and 105 boats (9 March) in the 250-berth small boat harbour. Our boat census does not include active boats in the river and those berthed in nearby docks outside the harbour basin. These figures show seasonal fluctuations of boat numbers in Port Alfred and suggest an increase in the number of new and transient boats in the Kowie River around the holiday season, coincident with the novel detection of *P. segnis* (this report). Finally, there is also the possibility that the specimen was passively transported over long distances as an adult, perhaps, an outcome of

anomalous current conditions. Should the specimen suddenly appear in temperate waters as an adult, it would probably experience a considerable physiological (thermal) barrier to survival and, eventually, die.

CONCLUSION

Invasions of marine and estuarine systems by true crabs (Decapoda, Brachyura) are a global concern, especially given the relatively high invasion success of crabs compared to other taxa (Por, 1978; Ambrogi, 2000; Brockerhoff & McLay, 2011; Tavares, 2011; Swart et al., 2018; González-Ortegón et al., 2020). In particular, alien crab species have been known to migrate across major biogeographic barriers either in response to climate warming (e.g., Antarctic invasions; Aronson et al., 2007) or via human mediation (e.g., fisheries activities, Lessepsian migration through the Suez Canal, ballast water discharge; Brockerhoff & McLay, 2011). Given its long and biogeographically varied coastline, South Africa is known to be susceptible to invasions by crab species and the novel detection of a live *Portunus segnis* outside its native range underscores the importance of screening for future invasions (e.g., Swart & Robinson, 2019). Importantly, however, this report highlights the additional need to screen for extralimital species, especially those breaching clearly defined biogeographic boundaries. For instance, crab species with disjunct distributions are often missed as aliens when they are detected in ‘gap’ region(s) sandwiched between their known ranges (Tavares, 2011), which could further distort our ability to document their spread and understand patterns of invasion by alien crabs. Alarming, range extensions of other marine taxa, including crabs, have been observed along the South African coast (e.g., Branch, 1984; Lamberth et al., 2011; Lloyd et al., 2012; Whitfield et al., 2016), which suggests that the appearance of *P. segnis* in temperate waters is part of an ongoing regional trend of range shifts — a trend that may be linked to climate change in southern Africa, such as ocean warming (Moloney et al., 2013; Potts et al., 2014; Kajee et al., 2018; Kelly et al., 2019). Our review of range-extending crabs demonstrates an overall pattern of poleward (in our case southwestward) shifts in distribution. Among the species with extralimital occurrences in southern Africa, *P. segnis* is the only one that is known to be spreading elsewhere in the world, namely, in the Mediterranean Sea, which underscores its potential for further spread in response to human-mediated climate change.

Following the detection of *P. segnis* (this report), surveillance of this species throughout the Agulhas ecoregion is recommended for rapid response actions (Lockwood et al., 2013). Surveillance efforts can also engage citizen scientists to assist in reporting suspected sightings of *P. segnis* and other alien crabs (i.e.,

early detection) and perhaps even document range expansion of established alien crab species (Delaney et al., 2008; Grason et al., 2018). Although the vectors and pathways of the spread of *P. segnis* in southern Africa remain unknown, we speculate that they are probably linked with maritime activities. The superficial resemblance between *P. segnis* (and other species in the *P. pelagicus* complex) and *Callinectes sapidus* may result in future misidentifications. For instance, records of *C. sapidus* as an alien species are confirmed for European, northern African, and Japanese waters (Doi et al., 2011; Shaiek et al., 2021), but its occurrence and native/alien status in Nigeria is uncertain and may even be a misidentification of a native congener (Jensen, 2010; Ololade et al., 2011). The potential for taxonomic misidentifications has implications to understanding the invasion histories (e.g., range shift patterns) of species and any ecological effects they may have on native organisms and communities.

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